Image-based Intrusion Detection System for GPS Spoofing Cyberattacks in Unmanned Aerial Vehicles

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***Abstract*—The operations of unmanned aerial vehicles (UAVs) are susceptible to cybersecurity risks, especially because of their strong reliance on the Global Positioning System (GPS) and radio frequency (RF) sensors. GPS and RF sensors are vulnerable to potential threats such as spoofing attacks that can cause the UAVs to behave erratically. Since these threats are widespread and potent, it is imperative to develop effective intrusion detection systems. In this paper, we present a deep learning-based methodology for detecting GPS spoofing cyberattacks. We combine convolutional neural networks with transfer learning to design a method that is fast, accurate, and general. The effectiveness of the proposed solution is demonstrated by extensive numerical experiments carried out using benchmark datasets. We achieved an accuracy of 100% within a running time of 120.64 s and a detection time of 2.035 s in the case of the training dataset. Further, using this trained model, we achieved an accuracy of 99.25% within a detection time of 2.721 s on an unseen dataset that was unrelated to the one used for training the model. Our results demonstrate that the proposed Image-based intrusion detection method outperforms the existing solutions while providing a general model for detecting cyberattacks included in unseen datasets.**

***Index Terms*—Unmanned aerial vehicles, CNN architectures, Transfer learning, Learning and loss curves**

I. Introduction

A. Background

Unmanned aerial vehicles (UAVs), commonly referred to as drones, are aircraft that fly automatically without any human pilot, crew, or passengers on board. UAVs were originally employed for military tasks, such as surveillance, bomb detection, delivery of armed payloads, and various missiles, such as AGM-114 Hellfire used by the United States [1]. In recent times, UAVs have also attracted significant interest from civilian industries for their potential applications for various civil requirements such as entertainment, inspection, surveillance, and mapping, assistance during natural disasters, and intelligent urban traffic control [2]. Further, according to some studies [3], [4], the market value of UAVs is expected to reach $1.85 billion by the year 2024.

Most civilian tasks rely heavily on the radio frequency (RF) band that enables data transmission, such as real-time video streaming, sensor data [5], and signals received from the Global Navigation Satellite System (GNSS) that provides positioning, navigation, and timing services [6]. Examples of a GNSS include Global Positioning System (GPS) owned by the United States, Global Navigation Satellite System (GLONASS) owned by Russia, Galileo Global Navigation Satellite System owned by the EU, and Quasi-Zenith Satellite System owned by Japan.

However, modern improvements in the connectivity of UAV networks for executing tasks have significantly increased the cybersecurity risks. This is due to the fact that UAVs primarily rely on GPS and RF sensors that are easily exposed to potential cybersecurity threats, such as spoofing attacks [7]. In a spoofing attack, counterfeit GPS signals control the command functions and steal sensitive data by replacing the GPS signals with a false signal that mimics the GPS format, causing the UAV to be set to a wrong trajectory [8]. There are several examples of spoofing attacks occurring in the real world. Researchers at the University of Texas took control of an $80 million yacht in the Mediterranean Sea by changing the trajectory of the yacht for about 100 m [9]. Iranian forces successfully spoofed a U.S. Lockheed Martin RQ-170 Sentinel UAV, forcing it to change its destination to Iran in December 2011 [10]. Researchers at Virginia Tech, China, and Microsoft changed the destination of an autonomous vehicle by manipulating the navigation system using their own-built $225 GPS spoofer in 2018 [11]. These studies were investigating the vulnerability of GPS systems to spoofing attacks by using software-defined radios (SDRs), which encompass many devices, such as High Amplitude Complex Keying Radio Frequency One (HackRF One), Great Scott Gadgets HackRF software-defined radio (HackRF), Universal Software Radio Peripheral (USRP), and Analog devices Active Learning Module - Pluto (ADALM-Pluto) [12]. SDR devices are a type of radio communication system that receive and generate signals using software by converting signals into RF signals (GPS signals) on a general-purpose computer instead of traditional hardware components like filters and amplifiers, making them more adaptable and reconfigurable for attackers compared with traditional hardware components [12]. To conduct a GPS spoofing attack by using an SDR device, the cyber-malicious agent (i) first identifies the authentic GPS signal that the UAV receives; (ii) uses a GPS signal simulator like the Global Position System–Software-Defined Radio simulator (GPS-SDR-SIM) [13] or the Global Navigation Satellite System–Software-Defined Radio Library (GNSS-SDRLIB) software [14] to parse the CSV file created by the cyber-malicious agent that holds the spoofed location and number of movements in the Earth-centered, Earth-fixed coordinate system (ECEF) format; (iii) employs the SDR device to convert the GPS baseband signal data streams created by the GPS signal simulator into GPS signals (the desired

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signal by the hacker); and (iv) transmits the spoofed GPS by the SDR to the UAV after adjusting the frequency and the signal power. As a result, the UAV will lose the authentic GPS signal that indicates the true location and time and will lock onto the spoofed GPS signal in 15 to 300 s depending on the embedded algorithms and the chosen spoofed GPS strategy. Therefore, GPS spoofing has severe effects on applications that depend on synchronization (accurate positioning and timing).

B. Related research, its limitations, and our contribution

In the context of UAV cyberattacks, previous studies have focused on conventional methods of signal processing and cryptography to analyze and defend against a spoofing GPS attack: (i) the cryptography technique involves encrypting the satellite codes received and sent by the GPS receiver. This technique requires a public key to decrypt the encrypted code; a digital signature to verify the authenticity and integrity of the code; a key management system to generate the public and private keys; and a secure communication protocol to grant the authenticity and integrity transmission of the encrypted and signed code by a digital signature between the satellites and the GPS receiver. However, this technique has a substantial cost: it adds complexity and may not be compatible with older versions of receivers and signals; (ii) the signal distortion detection technique tries to find any anomalies by monitoring the quality and characteristics of the GPS signal. The technique is implemented by measuring the strength of the signal and the frequency compared with GPS signals under normal conditions; however, it cannot detect complicated spoofing attacks that could seamlessly shift to the GPS receiver without any abrupt changes or errors; and (iii) the direction-of-arrival sensing technique uses multiple antennas to estimate the original GPS signal direction and compares it with the authentic GPS signal direction or the known direction of the satellites. The drawback of this technique is that it is affected by environmental factors that may block the GPS signals [15]–[17].

Kwon et al. [18] used the probability density function (PDF) to evaluate the accelerometer reading from an inertial measurement unit (IMU) to detect a GPS spoofing attack. The proposed method is carried out by monitoring the difference in the accelerometer readings under standard authentic GPS signals and spoofed GPS signals.

In [19], [20], the authors used the outputs of accelerometers that measure linear acceleration and the results of onboard gyroscopes that measure angular velocity or rotation rate, respectively. Both studies compare the outputs of IMU measurements that are computed from both the UAV and the GPS to determine whether the outputs are match. If the outputs of the IMU measurement do not match the GPS outputs, it indicates that the GPS is spoofed. However, IMU measurements are prone to various errors, such as bias and noise, that may affect the measurements of angular rate and acceleration, which are integrated into the inertial navigation system (INS). This error accumulation results in an inaccurate state of the UAV, making it hard to distinguish between standard authentic GPS signals and spoofed GPS signals [21].

Other studies [22], [23] have suggested that using artificial intelligence (AI) algorithms may have some advantages over all of the above techniques because the algorithms may gain experience from diverse spoofed GPS scenarios. In [24], the authors proposed a Long Short-Term Memory (LSTM) algorithm to detect GPS spoofing attacks by predicting Unmanned Aircraft System (UAS) paths and comparing them with the authentic GPS positioning signal. This comparison allows the algorithm to confirm whether the UAS is following an authentic or a spoofed trajectory. Although the detection accuracy of the proposed model reached 78%, the method is not applicable if the UAV shifts smoothly to the spoofed GPS with a small error deviation compared with the accepted threshold that the cyber-malicious agent has predefined.

Semanjski et al. [25] used a known supervised learning algorithm, C-Support Vector Machine (C-SVM), to detect GPS spoofing attacks depending on the features of the receiver clock that synchronizes with the satellites to determine the location of the UAV in real time. This allowed the authors to monitor the behavior of the receiver clock during both authentic and spoofed GPS signals. They claimed that the cross-validation accuracy of the C-SVM model and the overall accuracy were 97.8% on real-world data. However, they did not mention how they handled imbalanced datasets. Moreover, in their study, the GPS spoofing attack is an unusual occurrence, and thus, the number of normal samples is much higher than the malicious samples. As a result, since the C-SVM is very sensitive to imbalanced datasets, the model may have a high overall accuracy but perform inaccurately on minor classes.

In [26], the authors employed the Multilayer Perceptron model (MLP) to detect GPS spoofing attacks by using path loss measurements collected from nearby cellular base stations. Their model achieved an accuracy of 80% with one base station and 93% with three base stations. However, this method relies on the cellular network, which, itself, could be jammed, affecting the path loss measurements and leading to inaccurate detection. Moreover, UAVs are used to perform in high-risk environments, and there may be issues with cellular network connectivity in urban or dead zone areas.

Whelan et al. in [27] and [28], used anomaly detection models such as one-class support vector machine, local outlier factor, and autoencoder to detect GPS spoofing and jamming based on their proprietary dataset named UAV Attack Datasets. The models were trained on flight data only (no attacks occur) to recognize normal behavior, allowing the proposed models to detect any deviation to be flagged as a cyberattack. The performance of their models achieved macro-averaged F1-score of 90.57% and 94.3% for GPS spoofing and jamming respectively.

Thus, the existing traditional signature-based detection strategies and intrusion detection systems (IDSs) presented above have the following limitations: (i) many of these studies do not mention how to handle imbalanced datasets; (ii) most of the studies achieved a high accuracy without showing any learning or loss curves for training and validation; (iii) most of the studies did not test the proposed model with an unseen GPS spoofing dataset; and (iv) only a few studies considered

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the detection and execution time, although they are essential in cybersecurity, where time-sensitive activities must be carried out while detecting and thwarting UAV cyberattacks and the proposed models are computationally expensive. Therefore, it is crucial to develop a detection system to protect UAVs from spoofing attacks. The main contributions of this paper to bridge this gap in the previous researches are:

- We introduce an image-based intrusion detection framework that employs convolutional neural network architectures with transfer learning for detecting GPS spoofing attacks;
- We discuss the uses of different oversampling techniques, such as the Synthetic Minority Oversampling Technique with the Edited Nearest Neighbors and Adaptive Synthetic Sampling algorithm, and we determine the most suitable oversampling technique for the given datasets;
- We employ data transformation with pipeline optimization method to streamline the time taken for image loading and processing, harmonizing feature dimensions to address discrepancies in the number of features for training and unseen datasets using principal component analysis, and Bayesian optimization with tree Parzen estimator to reduce the overfitting;
- We show, for the first time, how a high-accuracy model that overfits can cause a problem in detecting cyberattacks by not generalizing properly, and propose a method to reduce overfitting and to generalize well to unseen dataset;
- We discuss the significance of illustrating the combination of the transfer learning, learning curves, and loss curves in one plot to illustrate the effectiveness of the chosen existing deep learning methods, and to evaluate the performance and the convergence of the proposed models.

To the best of our knowledge, no previous work has proposed such an image-based intrusion detection method that combines CNN architecture models with transfer learning to detect GPS spoofing attacks from a UAV attack dataset; illustrated a plot that combines learning and loss curves for training and validation before and after applying transfer learning; evaluated the proposed models with many different classification metrics; meticulously detailed the testing process; or rigorously evaluated the proposed models using unseen datasets. Developing a model to identify diverse GPS spoofing attacks poses a formidable challenge due to the various types of UAVs with different types of sensors, and communication protocols. This diversity in UAV types and features increases the complexity and intensifies the intrusion detection problem, resulting in a shortage of model performance due to the diverse set of features in UAVs that the model has not been trained on. Therefore, prior research works that have implemented machine learning approaches or proposed their algorithms and achieved high accuracies have often demonstrated their models only on specific dataset. However, the generalizability of these models to unseen datasets remains uncertain.

The rest of the paper is organized as follows. Section II explains the key concepts behind the implementation of the models, introduces the proposed framework, and provides a comprehensive description of the characteristics of the training and unseen datasets. Section III demonstrates the proposed model evaluation metrics, the step time, and the curves (learning and loss) for the best-performing proposed models. Section IV introduces and explains the model evaluation metrics for the best-fit models; presents the numerical results of the application of proposed models on an unseen dataset; and discusses the advantages of our framework and the challenges encountered during the experimentation. Section V concludes the paper.

II. Proposed framework

A. System Architecture

The objective of the present contribution is to develop an image-based intrusion detection method employing deep learning models with highly accurate forecasting information of GPS spoofing cyberattacks, obtained from *UAV Attack Datasets*[1]. The datasets consist of logs from benign flights and other flights that experience GPS spoofing attacks in two different formats: comma-separated values (CSV) and universal log (ULOG) files: they were created by Whelan et al. [8] and other authors from Ontario Tech University and the University of Tabuk.

In this experiment, the UAVs are equipped with an open-source flight controller named Pixhawk 4 flight controller [29] running the PX4 Autopilot software and launched in Shanghai, China, by the QGroundControl software that allows the user to execute the mission and monitor it in real-time [30]. During the flight, the UAVs receive authentic GPS signals from the Keysight EXG N5172B signal generator, while the agent simultaneously uses the GPS-SDR-SIM tools to generate fake GPS signals [31]. HackRF transmits the spoofed GPS signals with the same frequency band according to international regulations and a higher power than the authentic signal so that the UAV can lock onto the spoofed GPS. Moreover, to guarantee a successful hijack, the agent broadcasts white Gaussian noise on the GPS frequency band to jam the GPS signals in order to prevent the UAV from receiving authentic GPS signals. Further, each UAV flight has different flight dynamics and characteristics that may affect a model's detection of GPS cyberattacks. The generated datasets are thereby useful for evaluating the performance of the models under GPS spoofing and ping DoS attacks, because they contain both simulation and live flights of different types of UAVs, such as Holybro S500, Yuneec H480, DeltaQuad VTOL, Standard Tailsitter, Standard Plane, and 3DR IRIS [27], [28].

In this paper, we propose an image-based IDS model based on CNN architectures with transfer learning to detect GPS cyberattacks. Thus, the proposed deep learning model architecture in Fig. 1 is designed not only to achieve a high classification accuracy on the training dataset but also to generalize well to other datasets relevant to GPS attack contexts. Therefore, our IDS model is evaluated by using many different classification metrics and moreover, the model is

[1] https://https://ieee-dataport.org/open-access/uav-attack-dataset

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tested on an unseen dataset to ensure successful generalization on a new dataset. Additionally, we introduced a plot that combines learning and loss curves for training and validation before and after applying transfer learning, which will be presented later Section.

The proposed image-based IDS model consists of the following seven main stages:

1) **Data Collection**: Collecting network traffic datasets in one comma-separated values file;
2) **Data Preprocessing**: Removing duplicated rows, handling the missing values, converting raw data into a format that deep learning models can use for detection, reducing the number of features, and balancing the dataset;
3) **Data Representation**: Converting tabular data into three-dimensional array images and saving them as a PNG file;
4) **Data Splitting**: Splitting the train-test-validation split for the dataset as follows: An 80%-13.33%-6.67% train-validation-test split is used to generate a training set with 80% of the data samples, a validation set with 2/3 of the original 20% of the data samples, and a test set with 1/3 of the original 20% of the data samples;
5) **Data Augmentation**: This is applied randomly to each image in the training set in the following manner. The image is randomly flipped horizontally and/or vertically with a probability of 0.5; the image is randomly rotated up to 20% of its width and height; and the image is randomly zoomed in or out by up to 10%;
6) **Model Training**: Using training CNN architecture models, such as MobileNetV2, Xception, VGG16, VGG19, ResNet152, InceptionResNetV2, and InceptionV3, with transfer learning;
7) **Model detection**: Evaluating the detections for each CNN architecture model, and demonstrating the learning and loss curves for training and validation.

B. Data preprocessing

In order to obtain a high-performance IDS model for detecting GPS attacks, we used two datasets created by Whelan et al. [8] in our paper: (i) a training dataset containing about 85 sensor files with information on actuator controls, battery status, vehicle air data, and attitude, for five different UAVs; and (ii) an unseen dataset containing about 268 sensor files for one UAV named 3DR IRIS+. The differences between the datasets are presented in Table I.

However, the datasets have a few limitations:

- There are some duplicated rows that may cause biasing and reduce the standard deviation of the dataset. Therefore, it is necessary to remove these duplicated rows before conducting any further experiments.
- The datasets are vulnerable to the problem of high-class imbalance, which can lead to a low accuracy and a high false positive rate.
- Both datasets have a high number of outliers in some features, such as the *quaternion q[0,1,2,..]* that describes the position of the UAVs in space, *delta_q_reset[0]* that indicates the difference between previous and current quaternion, and *dist_bottom_valid*, which is the validation of the distance to the ground.
- The training and testing datasets have a high number of features, 85 and 268, respectively, which increases the model complexity and chances of overfitting [32].

TABLE I
DESCRIPTIVE STATISTICS OF THE UAV ATTACK AND UNSEEN DATASETS

Descriptive statistics	Training dataset	Unseen dataset
Number of UAVs	5 different UAVs	1 UAV
Number of features	85	268
Cyberattacks	GPS spoofing attack	GPS spoofing and Ping Dos attack
Mean	1.16003e+07	2.9569e+06
Median	2.9149e-02	3.6226e-04
Standard Deviation	1.5951e+08	8.2505e+07
Interquartile Range	1.2620	1.0
Variance	2.5445e+16	6.8072e+15

If the datasets are not correctly preprocessed to address and mitigate the limitations of the datasets, the input data might have a different format or shape than what is expected by the proposed models. The models might then take a long time to converge. Moreover, increased data complexity often leads to poor performances, with many models over- or under-fitting. Therefore, data preprocessing is one of the most important stages, having a significant impact on the performance of our proposed model. The preprocessing involves converting raw data into a format that deep learning algorithms can use to generate predictions, eliminate noise and outliers, and reduce the impact of irrelevant features. Hence, to create an IDS model with a high performance and an ability to generalize well to a new dataset, a few measures must be taken [33].

Raw datasets with different feature scales cause difficulties to the deep learning models during learning and result in slow, or no, convergence. Therefore, the dataset is normalized by $Quantile\ Transformer\ Normalization$ to ensure that all the numeric features have a comparable range of 0 to 1 [34], and the labels of the datasets are converted into numerical values between 0 and $n-1$ by using a $LabelEncoder$ from the sklearn library [35].

Because of the significant number of features and data logs in the dataset, it is worth reducing the dimensions of the data before training them with deep learning models to minimize the complexity and improve the performance of the model. Therefore, we implemented Principal Component Analysis (PCA), a dimensional reduction technique to reduce the number of features into principal components while keeping the most relevant data [36]. PCA can be performed mathematically in many ways; however, in our study, it was carried out by multiplying the standardized data with the matrix of the eigenvectors of the covariance matrix, resulting in a new dataset with fewer dimensions [37]. This involves (i) standardizing the range of the continuous initial variables to transform all the variables into the same scale to prevent a biased result; (ii) calculating the covariance matrix to address

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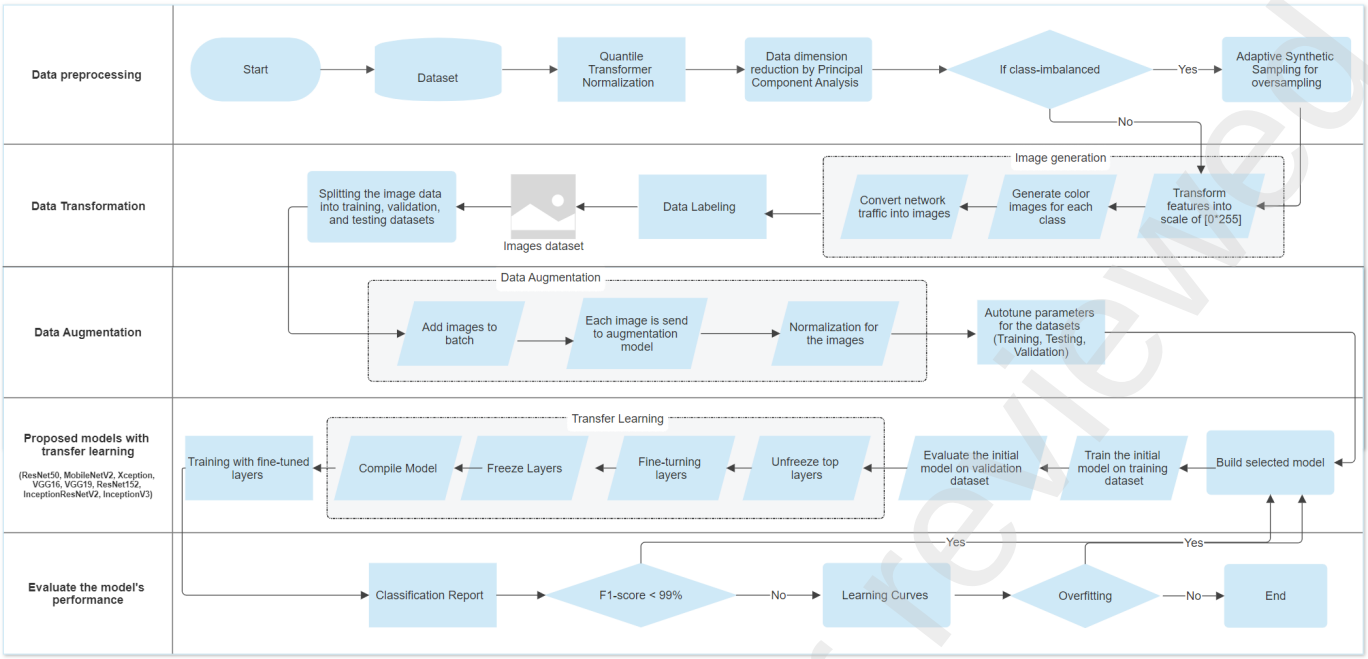

Fig. 1. Proposed image-based intrusion detection framework.

the relationship between the variables in the data as follows:

$$\frac{1}{n-1}(X-\bar{X})^T(X-\bar{X}) \tag{1}$$

where n is the number of samples, $\mathbf{X}$ is the data matrix, and $\bar{\boldsymbol{X}}$ is the mean vector; and (iii) calculating the eigenvectors with eigenvalues to determine the principal components based on the proportion of variance described by each component.

$$\Sigma v = \lambda v \tag{2}$$

where Eq. 1 is used to obtain eigenvalues (λ) and corresponding eigenvectors (v) for obtaining the desired reduced dimensionality [38]. These calculations are necessary for defining the number of principal components to transform the data into a new space [37]. In our studies, the number of principal components is visually illustrated by using a scree plot as shown in Fig. 2.

Imbalanced datasets are datasets with an uneven or unequal distribution of observations, indicating that certain class labels may have a large number of observations while others have fewer, as shown in Fig. 3. Imbalanced datasets are mainly solved by oversampling the minority class or undersampling the majority class, which can be achieved through techniques such as synthetic minority oversampling techniques (SMOTE), SMOTE with Edited Nearest Neighbors (SMOTE-ENN), and various other balancing methods available in scikit-learn for handling imbalanced datasets [39]. The core idea of the balancing methods is encapsulated in the following formula:

$$X_{\text{new}} = X_{\text{inst}} + \lambda \cdot (X_j - X_{\text{inst}}), \quad j = 1, 2, \ldots, k \tag{3}$$

where $\lambda \in [0, 1]$ is a random number and X_j is a randomly selected sample from the set $\{X_1, X_2, \ldots, X_k\}$ of k nearest neighbors of X_{ins} [40].

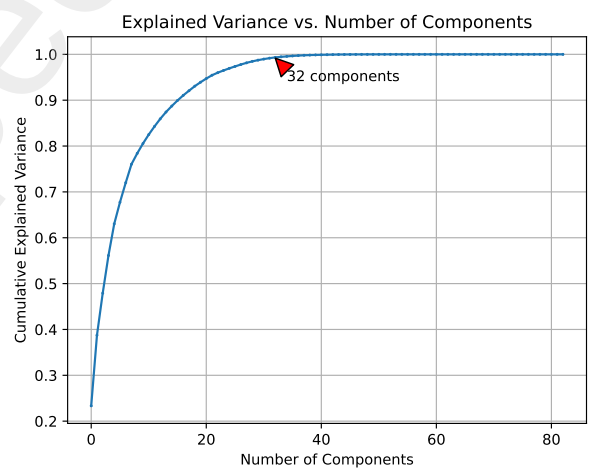

Fig. 2. Scree Plot of cumulative explained variance for principal component analysis.

SMOTE-ENN is one of the hybrid methods that combine both oversampling and undersampling techniques to address the imbalance class by creating new instances for the minority classes [39], then employs the edited closest neighbors (ENN) algorithm technique to eliminate noisy and borderline samples by comparing each synthetic sample with its k-nearest neighbors (KNN) [41]. Unlike random sampling, which simply duplicates the instances and may cause overfitting [42]. The combination of these two techniques aims to reduce the overfitting that might occur due to SMOTE and to remove some noisy samples by the (ENN) algorithm. However, removing noisy samples by using the ENN may also eliminate some informative samples, which results in a loss

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of valuable information and a low classification performance [43]. Therefore, we used the Adaptive Synthetic Sampling algorithm (ADASYN) method for oversampling imbalanced datasets, which addresses the limitations of the SMOTE and the SMOTE-ENN by the following method:

- Generating more synthetic examples, using a density distribution function in the difficult regions of the feature space, where the class distributions are sparse, dense, and harder to learn [44], [45];
- Performing better on highly imbalanced datasets, as shown in Fig. 3, by randomly and diversely choosing the nearest neighbors that have more influence on the synthetic sample, which can reduce the overfitting and improve the model generalization to an unseen dataset [46].

ADASYN algorithm key steps for synthetic sample generation can be represented as [44]:

$$s_i = x_i + (x_{zi} - x_i) \times \lambda \tag{5}$$

The process involves creating synthetic samples (s_i) based on adding a scaled difference between the original instance and its neighbor ($x_{zi} - x_i$), where (x_i) is an instance in the n dimensional feature space X (original instance), x_{zi} is a randomly chosen neighbor from the minority class (selected neighbor), and λ is a random number that acts as scaling factor that used to scale the difference between the original instance and its randomly neighbor number.

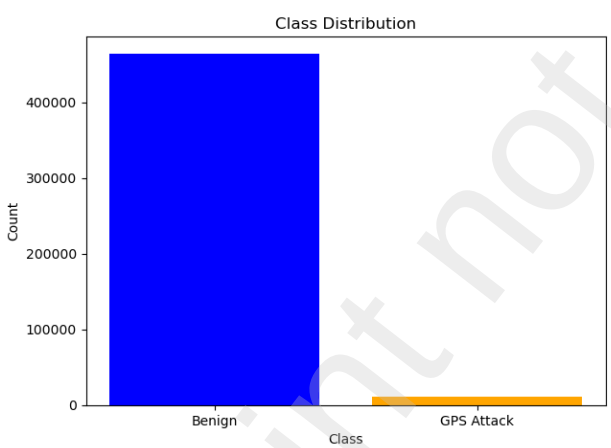

Fig. 3. Bar plot for the UAV dataset.

After preprocessing the dataset, we employed the network traffic image transformation technique, which is one of the data transformation methods used to convert the data into three channels of color images (red, green, and blue), because the CNN architecture models work effectively on images with an 8-bit depth (0 to 255) providing high-resolution data [47]. We carried out the network traffic image transformation by dividing the data samples into chunks based on the feature sizes. In our case, the training dataset has about 32323 features in an image that is transformed into ($32 \times 32 \times 3$), and each chunk has 32 consecutive data samples. The image generation for the Benign and GPS spoofing attacks is shown in Fig. 4.

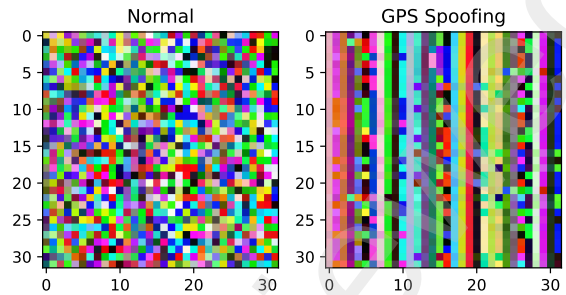

Fig. 4. Image generation for each class.

The datasets used in our paper were split by the holding split or train-validation-test split method, which involves splitting the datasets into three subsets: training, validation, and test. As mentioned in [48], the aim of using this method is to

- avoid the over-fitting issue where the model achieves a high accuracy only on the training dataset and a low accuracy on an unseen dataset. Therefore, by having a subset of the testing set, the model can be expected to perform well on an unseen dataset.
- avoid the under-fitting issue where the model learns poorly from the training dataset. By using a large subset of the training set, the model will learn from and capture the complexity of the dataset.
- achieve the best trade-off between bias and variance by using the subset of the validation set.

Data augmentation is one of the methods that are used to increase the training data by generating additional training samples so that the model can generalize well to unseen datasets (e.g., by employing the Keras Sequential API). The common data augmentation techniques used in deep learning are rotating, flipping, scaling, cropping, or adding noise to the existing training data to improve the model performance and reduce overfitting [49]. In this study, three random transformations are applied to the images: horizontal and vertical flipping, rotation, and zooming.

Loading datasets for training the deep learning models after the data preprocessing and augmentation can result in a high computational time. Therefore, the last critical step for reducing the computational time is to use data pipeline optimization, which involves various techniques like parallel processing, caching, prefetching, and compression. The aim of this step is to prepare and load the dataset without using a high capacity of the CPU and GPU resources while reducing the data transmission and the delay of memory access [50]. In our study, we reduced the computational time by using the prefetching technique that overlaps the data preprocessing and model training stages to speed up the training process. The aim of this step is to fetch the elements of the dataset in advance so that they are ready to be utilized by the next step [51].

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C. Transfer learning with Convolutional Neural Network architectures

CNNs are an artificial neural network (ANN) that have become a critical and popular architecture in deep learning particularly for image classification tasks as they can extract features from the input image and classify them into one of several predefined classes [52].

CNN architectures that are used for image classification tasks have several convolutional layers, followed by activation, spooling, dropout, and fully-connected layers. Convolutional layers are used to extract features from the input image by a set of filters or kernels. Each filter applies a convolution operation to create a feature map that highlights particular features in the input image to generate accurate predictions. By stacking multiple convolutional layers, the network can learn increasingly complex features from the input image. Activation layers, such as Rectified Linear Unit (ReLU) create nonlinearity in the network by selecting all the positive values in the feature map and setting all the negative values to zero. This allows the network to learn more features from images and improve its accuracy on image classification tasks. The pooling layer in the CNN is used to retain the most crucial data while downsampling the size of the feature created by the convolutional layers. Limiting the number of parameters while keeping the most crucial data aids in minimizing the computational complexity of the network and preventing overfitting. Dropout layers are used to prevent overfitting by randomly dropping out some of the neurons in the network during training. Fully-connected layers are used to connect all the features extracted by the previous layers and generate the final output.

The majority of CNN models contain three layers: a bottom layer that learns basic features such as edges; a middle layer that learns more intricate patterns such as shapes; and a top layer that learns high-level representations tailored to the dataset and task. By utilizing a pretrained CNN model, we can fine-tune only the top layers for our task while still leveraging the knowledge feature patterns learned by the bottom and intermediate layers. This technique is called transfer learning, which enables a new model to leverage the knowledge acquired by a pretrained model that has been trained on a large dataset [53]. This knowledge is gained by transferring the weights of the pretrained model to the new model trained on another dataset [54]. Moreover, by fine-tuning a pretrained model, we can achieve a high accuracy while using less computational resources. The fine-tuning phase converges more quickly than when training the CNN model from scratch and the risk of overfitting is reduced by preventing the model from remembering the training data. Therefore, in the context of CNN image classification, transfer learning has been successfully applied to many image processing tasks, because the captured patterns trained by the CNN model (pretrained model) on a large dataset are applicable to many different tasks.

ImageNet is a large-scale dataset that has more than one million labeled images belonging to 1000 different classes, and it is commonly used as a benchmark for image classification tasks. CNN architectures that are trained on the ImageNet dataset are considered some of the best models for image classification tasks due to their ability to learn from the complex features of raw image pixels and the use of transfer learning, which enables them to be fine-tuned for a wide range of tasks [53]. Therefore, we used CNN architectures, such as MobileNetV2, VGG16, VGG19, InceptionV3, InceptionResNetV2, Xception, and ResNet152, to train our training dataset and detect GPS spoofing by using transfer learning. MobileNetV2 is a lightweight architecture that has achieved an accuracy of 71.8% on the ImageNet dataset. It has 155 layers consisting of depthwise separable convolutions to reduce the number of parameters and linear bottlenecks to reduce the computational time and cost required to train the model; the last two layers are fully connected layers. VGG16 and VGG19 were developed by the Visual Geometry Group (VGG) at the University of Oxford. Both the models have achieved a top-5 error rate of 7.4% and 6.8% and an accuracy of 71.5% and 71.1%, respectively, on the ImageNet dataset. The key difference is that VGG19 has more filters in each layer, which makes it a more complex model. InceptionV3, developed by Google in 2015, has 48 convolutional layers and 3 fully connected layers, and it has achieved an accuracy of 78.0% on the ImageNet dataset. It uses the batch normalization technique that normalizes the activations of each layer to improve the accuracy of the model and reduces the overfitting issues of the model. The InceptionResNetV2 model was introduced by Google in 2016; it is an updated version of InceptionV3. The key difference is that InceptionV3 has 48 layers, while InceptionResNetV2 has 572 layers, which makes it possible to capture more complex features in the images. Xception is an extension of Inception that uses depthwise separable convolutions and a global average pooling layer at the end of the model to reduce overfitting and improve generalization. It has achieved an accuracy of 79.0% on the ImageNet dataset. ResNet152 is a deep architecture that has 152 layers and a total of 60.4 million parameters. It has achieved an accuracy of 78.6% on the ImageNet dataset and uses residual connections to address the vanishing gradient problem. The design of the bottleneck structures of those algorithms reduces the dimensionality of the feature maps, reduces the computational cost, and the risk of overfitting [55].

III. PERFORMANCE EVALUATION

The deep learning models proposed in this study were implemented on the UAV attacks dataset using Keras libraries and Scikit-learn. The models were trained by the computational resources provided by CSC - IT Center for Science Ltd., a nonprofit organization that offers IT services for research, education, and public administration in Finland. The computer configuration utilized for these experiments consists of Xeon Gold 6230 (2 x 20 cores @ 2,1 GHz) with a NVIDIA V100 (Tesla V100-SXM2 GPU) and 300 GB of memory.

To evaluate the performance of the proposed deep learning models on a highly imbalanced dataset for predicting GPS spoofing attacks, we focused on the following:

- We studied the following classification metrics to measure how well the proposed models will classify the spoofed

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GPS signal events: accuracy (ACC), which shows the accuracy of classification for each stage; detection rate (DR/recall), which is the ratio between the detected attack data and the total abnormal data; harmonic precision-recall mean (F1-score), which is used as a statistical measure to rate the model performance as it depends on two factors, precision (PRE) and DR/recall; receiver operating characteristic area under the curve (ROC AUC), which measures how the proposed models will perform to know the difference between authentic and spoofed GPS signals (positive and negative classes); precision-recall area under the curve (PR AUC), which measures the trade-off (balance) between precision and recall; balanced accuracy (BAC), which is the average recall of the benign and spoofed GPS signals and is a useful metric for imbalanced datasets; Matthews correlation coefficient (MCC), which is a statistical tool based on chi-square statistics used to measure the difference between the predicted values and the actual values; and Jaccard score (JSC), which measures the similarity between the labels, i.e., benign and spoofed GPS signals.

- We employed multiple loss function metrics, such as logarithmic loss function (Log Loss), which measures how good the performance of a model is by computing the differential between the predicted probabilities (the outputs of the model) and the actual values (true labels), and zero-one loss, which measures how many times the predicted label does not match the true label.
- We used the learning and loss curves for training and validation to monitor the performance of the model, i.e, whether it is overfitting, underfitting, or is a good fit. This allows us to track the deterioration of the learning performance of the proposed models.
- We used the confusion matrix to summarize the performance of the CNN models on a set of test and validation datasets. This is critical when dealing with imbalanced datasets and helps us to examine the model prediction errors by monitoring the false positive and false negative values.
- We computed both the running time of the models using the "time" library and the elapsed time per step (step time) to calculate how long the models need to process one batch during the testing phase.
- We evaluated the generalization of our models by testing them on unseen datasets.

The results of evaluating the proposed models for both the testing and validation datasets presented in Table II demonstrated that all the models achieved high accuracies. Even though the proposed models and the models in the literature achieved high accuracies, this does not necessarily mean such well-performing models that could generalize well to an unseen dataset. Therefore, we provided the learning and lose curves for our best-performing models to help us track the improvement or deterioration of the learning performance of the model by plotting both the training and validation loss and accuracy over epoch in one graph as shown in Figs. 5, 6 and 7.

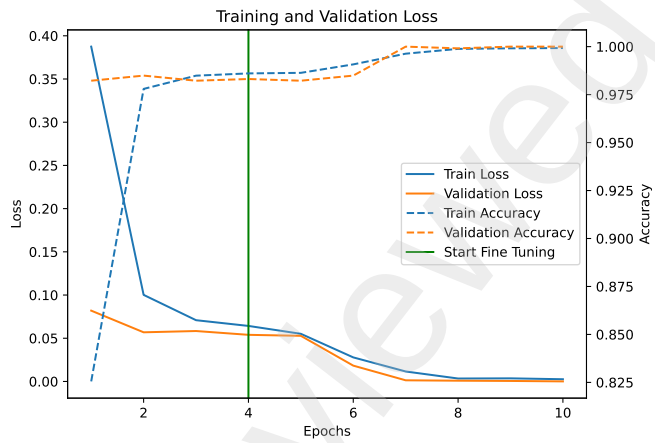

Fig. 5. Learning and loss curves for the Residual Networks model with 50 layers and a running time of 117.43 s.

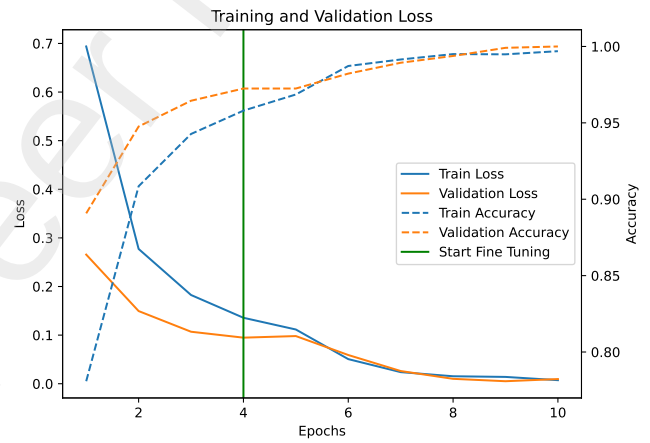

Fig. 6. Learning and loss curves for the Visual Geometry Group model with 19 layers and a running time of 120.64 s.

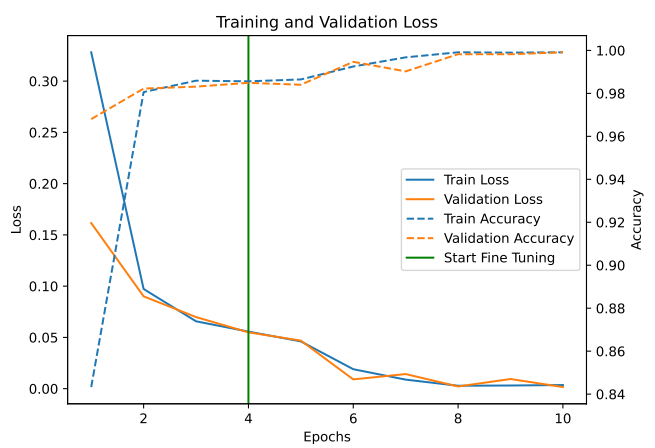

Fig. 7. Learning and loss curves for the Residual Network with 152 layers and a running time of 309 s.

IV. Results and Discussion

To construct a generalized robustness model to avoid overfitting issues, we trained the CNN models with transfer learning

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TABLE II
CLASSIFICATION REPORT FOR THE TESTING AND VALIDATION DATASETS OF THE UAV ATTACK DATASET

Classification report for the testing dataset with transfer learning								
Classification Metrics	**Models**							
	ResNet-50	MobileNet-v2	Xception	VGG-16	VGG-19	ResNet-152	Inception ResNet-v2	Inception-v3
Accuracy (%)	99.652	91.184	99.652	99.479	100	100	99.652	99.652
Precision (%)	99.654	93.127	99.654	99.483	100	100	99.654	99.654
Recall (%)	99.657	91.840	99.652	99.479	100	100	99.652	99.652
F1-score (%)	99.656	91.871	99.652	99.478	100	100	99.652	99.652
F1-score for each type of attack (%)	[99.690, 99.604]	[92.205, 91.438]	[99.705, 99.576]	[99.554, 99.373]	[100, 100]	[100, 100]	[99.693, 99.6]	[99.698, 99.590]
Confusion Matrix	[[322 0] [2 252]]	[[278 47] [0 251]]	[[339 0] [2 235]]	[[335 0] [3 238]]	[[319 0] [0 257]]	[[334 0] [0 242]]	[[325 0] [2 249]]	[[331 0] [2 243]]
ROC AUC (%)	99.606	92.769	99.578	99.377	100	100	99.601	99.591
PR AUC (%)	99.559	84.228	99.503	99.276	100	100	99.550	99.538
Balanced Accuracy (%)	99.606	92.769	99.578	99.377	100	100	99.601	99.591
Matthews Correlation Coefficient	0.99297	0.84880	0.99284	0.98933	1.0	1.0	1.0	0.99291
Jaccard Score	0.99212	0.84228	0.99156	0.98755	1.0	1.0	0.99203	0.99183
Log Loss	0.12515	2.94106	0.12515	0.18772	2.224e-16	2.220e-16	2.220e-16	0.12515
Zero-One Loss	0.00347	0.08159	0.00347	0.00520	0.0017	0.0	0.0	0.00347
Specificity	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Step time (s)	1.35	0.19	1.43	1.46	2.90	1.79	2.78	1.31
Classification Report for validation dataset with transfer learning								
Classification Metrics	**Models**							
	ResNet-50	MobileNet-v2	Xception	VGG-16	VGG-19	ResNet-152	Inception ResNet-v2	Inception-v3
Accuracy (%)	100	90.674	99.289	99.555	100	99.911	99.467	99.111
Precision (%)	100	92.234	99.298	99.555	100	99.911	99.472	99.125
Recall (%)	100	90.674	99.289	99.555	100	99.911	99.467	99.111
F1-score (%)	100	90.722	99.288	99.555	100	99.911	99.466	99.117
F1-score for each type of attack (%)	[100, 100]	[91.154, 90.140]	[99.383, 99.161]	[99.612, 99.480]	[100, 100]	[99.922, 99.895]	[99.536, 99.373]	[99.223, 98.962]
Confusion Matrix	[[642 0] [0 484]]	[[541 105] [0 480]]	[[645 0] [8 251]]	[[642 0] [5 479]]	[[657 0] [0 469]]	[[645 0] [1 480]]	[[644 0] [6 476]]	[[639 0] [10 477]]
ROC AUC (%)	100	91.873	99.168	99.483	100	99.896	99.377	98.973
PR AUC (%)	100	82.051	99.047	99.410	100	99.880	99.288	98.834
Balanced Accuracy (%)	100	91.873	99.168	100	99.483	99.896	99.377	98.973
Matthews Correlation Coefficient	1.0	0.82894	0.98555	0.99096	1.0	0.99818	0.98915	0.982025
Jaccard Score	1.0	0.820512	0.98336	0.98966	1.0	0.99792	0.98755	0.979466
Log Loss	2.2204e-16	3.36108	0.25608	0.16005	2.2204e-16	0.03201	0.19206	0.32010
Zero-One Loss	0.0	0.0932	0.007104	0.00444	0.00173	0.00088	0.00532	0.00888
Specificity	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Step Time (s)	1.34	1.19	2.42	3.42	4.53	3.77	9.129	1.30

to exploit the knowledge and features gained from a significant number of images and classes in the ImageNet classification dataset. We applied partial transfer learning for parameter transfer in our study by freezing the lower layers of the pretrained model and then fine-tuning the higher layers for our task.

After the partial transfer learning was implemented, the overall performance of the proposed models improved approximately by 0.8% to 2.08%. The partial transfer learning configuration and running time for the proposed models are presented in Table III.

The green vertical line in learning and loss curves serves as a visual indicator of the utilization of the partial transfer learning on the proposed models, preventing it from under- or overfitting. The curves before the green line demonstrate that the models are overfitting since the training undergoes without partial transfer learning as shown in learning and loss curves figures, and the confusion matrix indicates that the proposed models are not performing well in correctly identifying GPS cyberattacks, as shown in Table IV. Post the green line, the training incorporates partial transfer learning. Learning and loss curves figures aim to emphasize the influence of partial transfer learning on the model's overall performance as shown in Table II.

TABLE III
PARTIAL TRANSFER LEARNING CONFIGURATION AND RUNNING TIME FOR THE CNN MODELS

Models	**Total layers**	**Frozen layers**	**Unfrozen layers**	**RT (s)**	**DT (s)**
ResNet-50	175	1 to 150	151 to 175	117.43	1.951
MobileNet-v2	154	1 to 132	133 to 154	77.3	1.289
Xception	132	1 to 113	114 to 132	120.41	2.035
VGG-16	19	1 to 16	17 to 19	120.85	3.241
VGG-19	22	1 to 18	19 to 22	120.64	3.086
ResNet-152	515	1 to 443	444 to 515	309	5.102
Inception ResNet-v2	780	1 to 671	672 to 780	181.4	9.203
Inception-v3	311	1 to 267	268 to 311	106.4	1.964

Although we employed many classification metrics to eval-

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uate our proposed models, we focused on the F1-score metric and confusion matrix result values as the accuracy measures. In most real-world classification situations, the data might have an imbalanced class distribution, which can mislead other metrics, such as accuracy, but the F1-score metric and the confusion matrix provide a balanced evaluation. The F1-score considers both precision and recall metrics, thus making it a better indicator of a model's ability to generalize well to unseen data, and the confusion matrix provides a clear evaluation of how well the proposed models will perform in terms of correct or incorrect predictions.

TABLE IV
CLASSIFICATION REPORT FOR TESTING AND VALIDATION DATASETS BEFORE APPLYING PARTIAL TRANSFER LEARNING

Metrics for testing dataset	**Models**			
	ResNet-50	**ResNet-152**	**VGG-19**	**Inception-v3**
Acc (%)	98.611	99.652	98.784	98.437
PR (%)	98.643	99.654	98.810	98.479
Rec (%)	98.611	99.652	98.784	98.437
F1 (%)	98.607	99.652	98.782	98.433
F1\attack (%)	[98.809, 98.333]	[99.700, 99.586]	[98.940, 98.574]	[98.634, 98.174]
CM	[[332 0] [8 236]]	[[333 0] [2 241]]	[[327 0] [7 242]]	[[325 0] [9 242]]
Metrics for validation dataset	**Models**			
	ResNet-50	**ResNet-152**	**VGG-19**	**Inception-v3**
Acc (%)	98.579	98.401	97.957	98.490
PR (%)	98.613	98.445	98.026	98.529
Rec (%)	98.579	98.401	97.957	98.490
F1 (%)	98.575	98.397	97.949	98.486
F1\attack (%)	[98.765, 98.326]	[98.606, 99.125]	[98.284, 97.475]	[98.674, 98.245]
CM	[[640 0] [16 470]]	[[637 0] [18 471]]	[[659 0] [23 444]]	[[633 0] [17 476]]

The evaluation of the proposed models for both the testing and validation datasets presented in Table II demonstrated that all the models achieved high accuracies, and the best models for the UAV attack dataset are the VGG-19 and ResNet-152 models for the following reasons:

- The VGG-19 and ResNet-152 models successfully identified a large number of actual GPS attacks that occurred (positive cases). The true positives in the confusion metrics indicated that the models detected 319 and 334 actual cyberattacks for the testing dataset, and 657 and 645 for the validation dataset respectively. On the other hand, the confusion metrics for the other models, such as VGG-16 and Inception-v3, indicated that the models failed to detect some GPS spoofing attacks.
- The training logs indicate that the proposed models have low training and validation losses of 0.0027 and 0.0032, respectively, for the VGG-19 model, and low training and validation losses of 0.0019 and 0.0016, respectively, for the ResNet-152 model. These results suggest that both models perform well on the training and validation datasets.
- The Matthews correlation coefficient and the loss function metrics, such as log loss and zero-one, indicate that the model achieved perfect prediction by learning the underlying patterns and could generalize well to a different dataset.
- The learning and loss curves for both models indicate that the models are not overfitting, as shown in Figs. 6 and 7.

Achieving a high performance rate as shown in the classification report in Table II is a desirable outcome, but it does not indicate a complete evaluation of the model performance. A comprehensive evaluation involves considering the ability of the model to generalize well and to identify complex patterns and features of different underrepresented attacks, such as GPS spoofing attacks, apart from the normal network behavior (benign traffic). As a result, we also tested our models with an unseen dataset that contains 268 features for one UAV named 3DR IRIS+. The differences between the training and the unseen datasets are shown in Table I.

The evaluation of the proposed models on the unseen datasets, shown in Table V, demonstrates that the VGG-19 and ResNet-152 models achieved the highest accuracies and F1 scores. However, the confusion metrics show that the models failed to detect some instances (cyberattacks) that occurred. The reason behind this is that these instances were related to the Ping DoS attacks, and our models were designed to detect GPS spoofing attacks.

TABLE V
CLASSIFICATION REPORT FOR THE PROPOSED MODELS ON UNSEEN DATASETS

Metrics for unseen dataset	**Models**			
	ResNet-50	**ResNet-152**	**VGG-19**	**Inception-v3**
Acc (%)	75.250	99.0	99.25	46.437
PR (%)	83.673	98.018	98.260	21.564
Rec (%)	75.250	99.0	99.25	46.437
F1 (%)	74.416	98.999	99.249	29.452
F1\attack (%)	[71.345, 78.217]	[99.083, 98.899]	[99.326, 99.153]	[63.422, 0.0]
MCC	0.59248	0.98003	0.98491	0.0
Jacc	0.64227	0.97823	0.98321	0.0
Log Loss	8.92080	0.36043	0.27032	19.305
ZOL	0.24750	0.01000	0.00749	0.53562
CM	[[493 392] [4 711]]	[[865 0] [16 719]]	[[885 0] [12 703]]	[[743 0] [857 0]]

Although, the learning curve for ResNet-50 also revealed that the model did not exhibit overfitting as shown in Fig. 5, the model performance exhibited suboptimal performance on unseen datasets. This is due to the fact that ResNet-152 has more layers, parameters (has 152 layers and 60 million parameters), and more bottleneck structure in each residual block compared to ResNet-50 (50 layers and 25 million parameters) [56], which allows the model to generalize well to a new dataset in terms of domain or distribution [55]. However, the inception-v3 model showed a poor performance on the unseen dataset with an accuracy of 46.437% and an F1-score of 29.452%. This is due to the fact that the learning and loss curves demonstrated that the model is overfitting as shown in Fig. 8, although it achieved a high accuracy of 99.6% on the testing dataset as shown in Table II. Therefore, it is critical to

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monitor the learning and loss curves even if the model achieves high evaluation metrics on the training dataset.

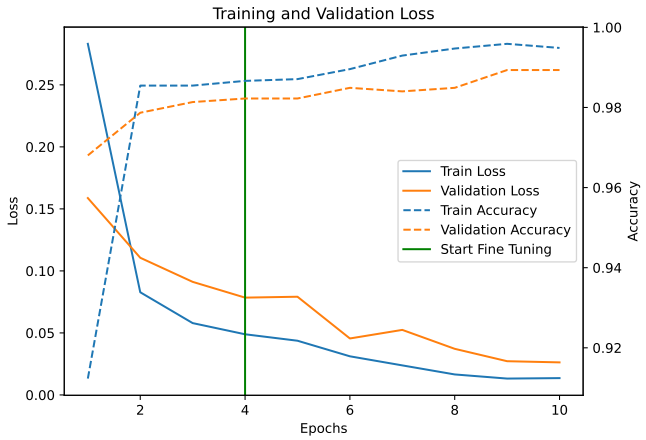

Fig. 8. Learning and loss curves for the Inception-Residual Version 3 model with a running time of 106.4 s demonstrated that the model is overfitting.

As a result, the parameters for Inception-v3 are tuned using a commonly used approach, the Bayesian optimization technique with Tree-structured Parzen Estimator (BO-TPE) algorithm in the Hyperopt library because of its sample efficiency [57]. Since there are many important hyperparameters in terms of accuracy, speed, and handling false negatives and false positives to prevent the overfitting, the chosen parameters for the Inception-v3 model are shown in Table VI. After implementing Hyperopt (BO-TPE), the performance of inception-v3 improved by approximately 0.35%, and the model is a good fit, as shown in Fig. 9.

TABLE VI
HYPERPARAMETER (BO-TPE) OPTIMIZATION CONFIGURATION FOR INCEPTION-RESIDUAL VERSION 3 MODEL

Hyperparameter	Search Range	Optimal Value
Frozen	[1, 780]	50
Epochs	[1, 20]	7
Patience	[2, 5]	4
Learning rate	[0.00003, 0.00009]	8e-05
Dropout	[0.2, 0.5]	0.4
Weight decay	[1e-9, 1e-3]	5.578e4
Optimizer	[Adam, SGD, RMSprop]	Adam
RT (s)	2184	

The learning and loss curve for the inception-v3 model in Fig. 8 demonstrated that the generalization gap between the validation and training losses indicates a lack of convergence. Therefore, to improve the learning and loss curve, we added the regularization parameter 'weight decay' into the hyperparameter tuning process using the BO-TPE algorithm in Hyperopt as shown in Table VI. As a result, the generalization gap has decreased as shown in Fig. 9 compared with Fig. 8.

In our study, we creatively used and combined existing deep learning approaches to effectively detect GPS spoofing cyberattacks with high accuracy and a low execution time, as shown in Table II, V, and VII. Further, to ensure a fair and valid comparison with other methods, we compared our performance method with different methods by using the same dataset for GPS spoofing attack and employed the same evaluation metrics as in Whelan et al. [28], including, Precision, Recall, and F1-score as shown in Table VIII.

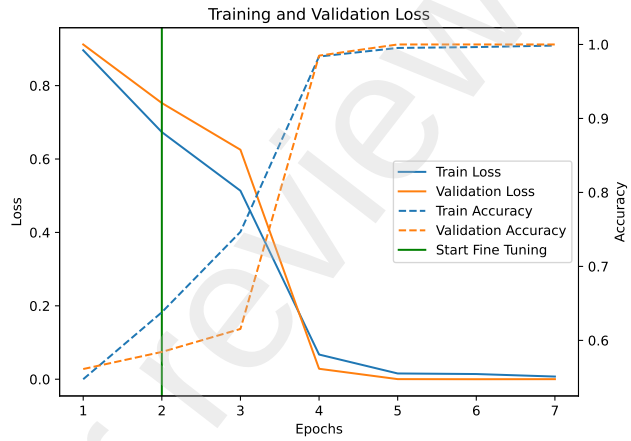

Fig. 9. Learning and loss curves for the Inception-Residual Version 3 model with Hyperopt and a running time of 75.1 s.

TABLE VII
CLASSIFICATION REPORT FOR INCEPTION-RESIDUAL VERSION 3 ON TRAINING AND UNSEEN DATASETS: HYPEROPT (BO-TPE) OPTIMIZATION

Classification Metrics	Training Dataset		New dataset (unseen)	
	Testing	Validation	Testing	Validation
Acc (%)	100	100	99.062	98.562
F1 (%)	100	100	99.061	98.556
F1\attack (%)	[100, 100]	[100, 100]	[99.168, 98.926]	[98.869, 98,025]
CM	[[336 0] [0 240]]	[[648 0] [0 478]]	[[894 0] [15 691]]	[[1006 0] [23 571]]
ST (s)	1.31	1.30	No training	
RT (s)	75.1			
DT (s)	1.957		3.703	

TABLE VIII
COMPARISON BETWEEN THE LATEST PROPOSED BASED ON UAV ATTACK DATASET.

Model	Classes	Precision	Recall	F1
Autoencoder [28]	Benign	99.36	94.81	97.03
	Malicious	74.72	96.18	84.10
Inception-v3 + Hyperopt	Benign	100	100	100
	Malicious	100	100	100

In this paper, we identified the following limitations and challenges for the problem investigated in this paper:

- Selecting an appropriate technique to build a high-performance model in terms of high accuracy without overfitting issues, low running and detection time, and generalizing well to unseen datasets is a formidable challenge. The fact that there are a multitude of models, diverse feature reduction methods, such as random forest regressor (RFR) and PCA, and different methods of data

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normalization techniques, such as min-max, Z-score, and Quantile transformer normalization, makes the decision-making process complex and intricate.
- Transfer learning can lead the proposed models to a poor performance or overfitting, if we fine-tune many layers, and using a complex pretrained model may require high computational resources if the features learned by the pretrained model are irrelevant for detecting a GPS spoofing attack.

Our proposed Image-based Intrusion Detection System for detecting GPS spoofing cyberattacks attempts to overcome these limitations. We observed several trends and patterns that demonstrate the effectiveness of our approach:

- In addition to the normal network behavior (benign traffic), the proposed models also successfully identified complex patterns and features of GPS spoofing attacks by using partial transfer learning techniques to reduce overfitting and minimize the false negatives and positives.
- The proposed model was trained on dimensionality-reduced features that exhibit strong correlations and are comparable with various GPS cyberattack datasets. This indicates the robustness of the model and its potential for real-world applications.
- The learning and loss curves showed that the training score and the validation score converged to a similar value as the number of training samples increased. This indicates that the proposed model has a good generalization ability and could perform well on unseen data.

V. Conclusion

UAVs are attracting significant interest among both military and civilian industries due to their ability to perform critical tasks in high-risk environments. At the same time, cyberattacks such as GPS spoofing attacks are also increasing significantly, constituting a serious threat to human lives. In this paper, we have proposed an Image-based intrusion detection system for detecting GPS spoofing attacks based on CNN architecture models with transfer learning. Our method was able to successfully learn and identify normal sensor values based on historical flight logs. The proposed models were evaluated on imbalanced datasets that include many different UAVs with different flight dynamics and characteristics affecting the GPS receiver's performance. The numerical results demonstrated that the selected methods enhance the performance of the model, leading to improved detection capabilities when tested on unseen datasets. The experimental results showed that the Visual Geometry Group 19-layer model and the optimized Inception version 3 model using the tree-structured Parzen Estimator algorithm with transfer learning achieved a high accuracy of 100%. Remarkably, the Visual Geometry Group 19-layer model and the optimized Inception version 3 also achieved a high accuracy of 99.2% and 99.06% on the unseen dataset respectively. Further, the model results demonstrated the feasibility of the proposed models in real-time data.

In the future, we plan to focus on different datasets and applications using different approaches such as an online adaptive model that can update itself on new available datasets. This will enable the model to remain accurate and relevant to new datasets.

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